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Computer security wk4

1. Scenario Analysis: Email Signing Using Cryptography

A company uses email signing to protect important messages sent between employees, customers, and business partners. When an employee sends an email, a digital signature is created using the sender's private key. The recipient can use the sender's public key to verify the signature. This process confirms that the email was actually sent by the claimed sender and that the message has not been changed during transmission. For example, a manager may send an employee an email containing an important financial report. The digital signature allows the employee to confirm that the report came from the manager and was not modified by an attacker. Cryptography therefore provides authentication, integrity, and non-repudiation. If someone attempts to change the email content, the signature verification will fail. Email signing is especially useful for organizations because it reduces the risk of impersonation, tampering, and fraudulent communication.

2. Concept Research: RSA Cryptography

RSA (Rivest–Shamir–Adleman) is a widely used public-key cryptography algorithm that helps protect digital information. It uses two related keys: a public key, which can be shared with others, and a private key, which must be kept secret. RSA can be used to encrypt information and create digital signatures. For example, a person can use a recipient's public key to encrypt sensitive information, which can then be decrypted using the recipient's private key. RSA also supports digital signatures, helping verify the sender's identity and ensure that information has not been changed. Its role is important in secure communication, authentication, and data protection.

3. Tool Practice: OpenSSL File Encryption Reflection

I used OpenSSL to encrypt a file and protect its contents from unauthorized access. The encryption process converted the readable file into encrypted data that could not be easily understood without the correct password or decryption key. This practical activity helped me understand how cryptography can be applied to protect sensitive information. I learned that encryption is useful for securing files when they are stored or transferred between users. I also discovered the importance of using strong passwords and keeping encryption keys private. Overall, using OpenSSL improved my understanding of confidentiality, data protection, and the practical role of cryptography in cybersecurity.

4. Diagram Design: Encryption Workflow

The encryption workflow diagram shows how information is protected using cryptography. First, the sender starts with an original file or message that needs protection. The file is then encrypted using an encryption algorithm and the recipient's public key. This changes the readable information into ciphertext that cannot be understood without the correct key. The encrypted file can safely be stored or transmitted. The recipient uses their private key to decrypt the ciphertext and recover the original file. The diagram also shows that public keys can be shared, while private keys must remain secret. This process helps provide confidentiality and protect sensitive information.

